

## Lab 6

**NO** late submission. Grade = 0 after the deadline.

**Due:** 13:50h 05/11/2025

Submission:

- **Lab#\_YOURNAME.ipynb**
- Must include all data file (**.csv**,...) and **.ipynb** in your submission
- **.csv** and **.ipynb** must be in the **same** folder (pd.read\_csv('data.csv'))

No use of LOCAL libraries, NO local directory for libraries

In this Lab, you will investigate different stock performances in **August 2025**. A **ticker** of a stock is its ticker symbol, which is a unique grouping of letters that identifies a specific company. For example, Apple has a ticker symbol of "AAPL".

For the Lab, the stocks (tickers) are NVIDIA (**NVDA**), Apple (**AAPL**), Microsoft (**MSFT**).

You will need to use:

**import yfinance as yf**

**import numpy as np**

**import matplotlib.pyplot as plt**

You will download the data as follows:

```
download = yf.download(ticker, start=start_date, end=end_date)
data = np.array(download)
dates = np.array(download.index.values)
```

You only need to use the **first** column of **data**.

You need to process the **dates** (numpy datetimes) so that they can be displayed on charts.

Each ticker should have 2 charts: one **line** chart shows the stock prices in August and the other **bar** chart shows the daily change. Note that if the change is positive, it should be **green**, otherwise it should be **red**. Hint: you can use **np.diff()**.

Finally, in order to show 3 charts in 1 graph, their stock prices should be scaled to [0-1]. The marker and color should be unique for each ticker in this chart.

**Total charts required for this Lab: 7 charts**

Sample outputs:

